

Malaviya National Institute of Technology, Jaipur

Mid Term Exam September-2025
Programming with Python (22CST101)

Duration: 1.5 hrs.

Max. Marks: 20

Q1. Answer the following Questions

- Convert $(153.513)_{10}$ to $(\quad)_8$
- Perform the following conversions. (i) $(234)_{10} = (?)_2$ (ii) $(73)_{10} = (1021)_2$
- Perform $12 + (-2)$ using 2's complement arithmetic.

(1 mark)

(2 marks)

(2 marks)

Q2. Answer the following Questions.

a) Which of the following are valid/invalid variable names in Python?

- my_var
- 2ndValue
- value\$
- result

(1 mark)

b) What will be the output of the following code?

```
x = 5
y = "5"
print(x + int(y))
```

(1 mark)

c) Identify the data types of the following literals:

- 3.14
- "Python"
- True

(1 mark)

Q3. Write a program that reads in the radius of a circle and prints the circle's diameter ($2r$), circumference ($2\pi r$) and area (πr^2). Use the constant value 3.14159 for π . Do these calculations in output statements.

(3 marks)

Q4. a) Write the output of the following code snippet.

(2 marks)

#A <pre>x = 5 y = 2 print(x ** y // y + x * y)</pre>	#B <pre>a, b = 7, 3 print((a & b) + (a b) - (a ^ b))</pre>
#C <pre>p = 10 q = 4 r = p >> 2 s = q << 1 print(r, s)</pre>	#D <pre>x = 2 y = 3 z = 4 print(x ** y ** z)</pre>

b) Evaluate the following expression: $(a * b // c - d + e / f)$ or False

Note: $(a, b, c, d, e, f) = (2, 4, 3, 7, 12, 7)$

(2 marks)

Q5. A mobile service provider charges its customers based on slab-wise data usage as follows:

(5 marks)

- Up to 2 GB: Free (₹0 per GB)
- Next 3 GB (from >2 GB to 5 GB): ₹10 per GB
- Beyond 5 GB: ₹20 per GB

In addition, every customer pays a base monthly plan of ₹200.

Write a Python program using an if-elif-else ladder to calculate the total bill amount for a customer based on data usage (in GB, which may be fractional).

else:

Print("Outer loop completed")



Malaviya National Institute of Technology Jaipur
Programming with Python (22CST101)
Mid-Term Examination

Date: February 21, 2025

Timing: 3:00 PM - 4:30 PM

Max marks: 30

Attempt all questions. Questions 1 – 5 carry 6 marks each.

1. (a) (3 points) Perform the following binary operations:
1. Convert the decimal number 57 to binary.
 2. Compute $1011_2 + 1101_2$ in binary.
 3. Perform a right shift operation on 00110100_2 by 2 positions.

- (b) (3 points) You are given the following Python code:

```
1 x = 15
2 while x > 0:
3     if x % 7 == 0:
4         break
5     elif x % 3 == 0:
6         x -= 4
7         continue
8     x -= 2
9 print(x)
```

1. Predict the output of the program.
 2. Modify the loop so that it prints all values of 'x' before exiting.
2. (a) (2 points) Identify the errors in the following Python code and correct them:

```
1 x = input("Enter a number:")
2 if x % 2 == 0
3     print("Even number")
4 else:
5     print("Odd number")
```

- (b) (4 points) Given a Python expression:

```
1 y = 10
2 z = 0
3 for i in range(1, y):
4     z += i * (y % i)
```

1. Predict the output of the given code.
 2. Modify the loop logic to calculate the sum of divisors of 'y' instead.
3. (a) (3 points) Consider the following Python program:

```
1 x = 5
2 while x > 0:
3     print(x, end=" ")
4     x -= 2
5 else:
6     print("Loop completed")
```

What will be the output? Explain how the 'while' loop and 'else' statement work in this case.

- (b) (3 points) Write a Python program to print numbers from 1 to 50 but skip multiples of 5 using the 'continue' statement.

4. (a) (3 points) Write a Python program that takes a string input from the user and reverses it without using built-in functions like '`[::-1]`' or '`reversed()`'. Use a loop to construct the reversed string and print the result. Test the program with the input: "Python".
- (b) (3 points) Given a list of numbers, write a Python program that finds and prints the largest number without using the '`max()`' function. Use a loop to solve the problem.
5. (a) (3 points) A multinational company decides to give a bonus to all its employees on Holi.
- A 5% bonus on salary is given to male employees.
 - A 7% bonus on salary is given to female employees.
 - If the salary of an employee is less than Rs. 20000, they receive an extra 3% bonus on salary.

Write a Python program to calculate the total bonus for an employee and display the final salary after adding the bonus.

Note: Your program should take the salary and sex (M/F) of the employee as input from the user.

- (b) (3 points) Write a Python program to generate the following pattern using nested loops:

```
1
1 2
1 2 3
1 2 3 4
```

- The program should take a single integer input n and print n rows of the pattern.

Example: If $n = 3$, output should be:

```
1
1 2
1 2 3
```



Malaviya National Institute of Technology Jaipur
Programming with Python (22CST101)

Mid Term Examination

Date: September 23, 2024

Timing: 3:00 to 4:30 PM

Max marks: 30

Attempt all the questions. Note: Questions 1 – 5 carry 6 marks each.

1. (a) Write a Python code using a 'for' loop that prints all the even numbers between 1 and 10. (4)

- (b) What will be the output of the following Python code? (2)

```
num = 7
if num % 2 == 0:
    print("Even")
elif num % 3 == 0:
    print("Divisible by 3")
else:
    print("Odd")
```

2. Write a Python script to compute the total amount payable after applying a discount, based on the user's initial purchase amount in INR.

Discount Rules:

- No Discount: For purchases less than 100.
- 5% Discount: For purchases from 100 to 500.
- 10% Discount: For purchases above 500.

Expected Outputs:

For a user input of 450, the output should be:

Original Amount: 450
Discount Rate: 5%
Discount Amount: 22.5
Total Amount Payable: 427.5

OR

Write a Python program using a while loop to calculate the sum of the first 10 natural numbers (Sum of 1 to 10. Output should be 55).

(6)

3. (a) Write a Python program that takes three integer inputs from the user, calculates the average of these integers, and prints the result as a float. (3)
- (b) Write a Python program that converts an integer representing seconds into hours, minutes, and seconds. For example, 3665 seconds should be displayed as 1 hour, 1 minute, and 5 seconds. (3)
4. (a) Perform the following logical shift operations to find the value of X (unsigned): (3)
- Given the expression $X \ll 3 = 8$ in binary, which is 00001000, where $\ll$ represents the left shift operation, find the value of X .
 - Given the binary number 00010000, perform a right shift operation by 4 positions ($00010000 \gg 4 = X$) to find the value of X .
- (b) What will be the output in the following cases, considering operator precedence and associativity? (3)
- [Each part carries 1 mark]*
- $3 * 4 // 6 + 5 - 3 * 7$
 - $3 * (4 // 6) - 3$
 - $(3 ** 4) ** 2 // 4$
5. a) Perform the following binary addition:

$$1011_2 + 1101_2$$

- Convert the decimal number 45 to binary.
- Convert the hexadecimal number A9 to binary.
- Which of the following are valid Python variable names? If invalid, explain why.
 - 'my_var'
 - '2nd.variable'
 - '_data123'
 - 'class'
 - 'firstName'
 - '@name'
 - 'employee_id'
 - 'True'

Best wishes

$$(1.5 * 4 = 6)$$

2023UCHI333



Malaviya National Institute of Technology Jaipur
Mid Term Examination 2023-24
B.Tech (Semester-II)

Subject: Programming with Python

Dated: February 26, 2024

Time: 16:15 - 17:45 Hrs

Subject Code: 22CST101

Full Marks: 30

Figures at the right margin indicate marks. All parts of a question must be answered in sequence.

1. Answer the following questions

- (a) You are given the marks obtained by a student in the Quiz test (out of 20), MTE (out of 30), and ETE (out of 50) for the "Programming with Python" subject. Your task is to design a flowchart to assign a grade to the student. [2]
- (b) Convert the followings numbers (i) $(9C.EF)_{16} = (?)_8$ (ii) $(254.76)_{10} = (?)_{16}$. [3]
- (c) Perform the following binary arithmetic operations: (i) $(5)_{10} - (12)_{10}$ (ii) $(82)_{10} + (-56)_{10}$ using 2's complement. [3]
- (d) For a given binary number $(0110011)_2$, perform 2-bit left shift and right shift operations, and determine the decimal equivalent of the results. 204, 12 [2]

2. Find out the outputs/errors (if any) of the following Python code fragment

[1×6=6]

(a) <pre> s1="MNIT Jaipur" s2=s1 if "n"==s[1]: print("Good") elif s2 is s1: print("Better") else: print("Best") </pre> <i>s is not defined</i>	(b) <pre> a=None b=5 if not a: print("True") if b: print("Hello") else: print("False") if (0): print("Hi") </pre> <i>Syntax error</i>
(c) <pre> i=1 while i<10: if (i%3==0): break print(i) i=i*2 </pre> <i>1, 2, 4, 8</i>	(d) <pre> numbers=[65,32,-3,32,43,32,2,-5,64] i=0 for n in numbers: if 2**i==n: print(n) i=i+1 </pre> <i># 32</i>
(e) <pre> a=3 (011) b=2 (010) print(a+b*a**b-a) print(a b,a^b) </pre> <i>18, 3.1</i>	(f) <pre> print(int(6 is 6.0) * 3 + 4 % 5) print(str(100)+int(100.25)) </pre> <i># Error</i>

print (int(6==6.0)) # 1

3. Write the Python code for the following

- (a) A multinational company decides to give bonus to all its employees on Holi. A 5% bonus on salary is given to the male employees and 10% bonus on salary to the female workers. If the salary of the employee is less than Rs. 20000, the employee gets an extra 3% bonus on salary. Write a program to calculate the bonus that has to be given to the employee and display the salary that the employee will get. (Note: Your program should take the salary of the employee and sex of the employee as input from the user.) [4]
- (b) You are given two positive integers *start* and *end*. Your task is to write a program to print all prime numbers in the interval [*start*, *end*]. (Note: The *start* and *end* are user-entered numbers) [4]
- (c) Write a program that prompts user to enter his/her first name and last name and then displays a message *Congratulations! 'Firstname Lastname'*, where Firstname and Lastname are the names entered by the user. [2]
- (d) Write a program to print the following pattern. [4]

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```